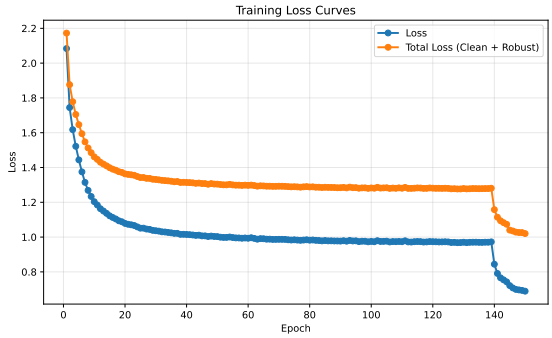
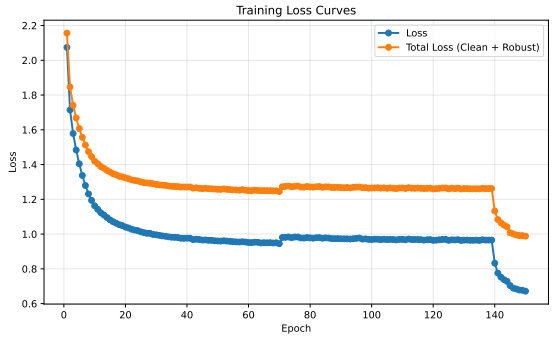



(a) TRADES




(b) TRIX

Figure 1: Comparison of training loss curves of TRADES and TRIX. Both methods appear stable during training and converge well.

Table 1: Comparison of robust accuracy on the CIFAR-10 validation set using a **ResNet-18** model under PGD-100 attacks with step size 2/255 at perturbation budgets $\epsilon = 16/255$ and 8/255. Results are reported as mean ± standard deviation for both average (AvgR) and worst-case (WstR) class accuracies across five random seeds.

| | $\epsilon = 16/255$ | | $\epsilon = 8/255$ | |
|---|---|---|---|---|
| Method | AvgR ↑ | WstR ↑ | AvgR ↑ | WstR ↑ |
| TRIX | **18.37 ± 0.27** | **7.22 ± 0.99** | **51.22 ± 0.18** | **35.26 ± 1.17** |
| TRIX (EMA, $\alpha = 0.95$) | 16.17 ± 0.22 | 6.08 ± 0.50 | 49.87 ± 0.29 | 34.86 ± 1.36 |
| CODAT | 15.98 ± 0.30 | 6.15 ± 0.77 | 49.80 ± 0.80 | 33.03 ± 1.87 |
| HAM | 16.20 ± 0.44 | 5.85 ± 1.02 | 51.05 ± 0.45 | 32.40 ± 1.71 |

Table 2: Performance comparison of TRIX and its EMA-based variant against CODAT (DR0-based reweighting; *Complex & Intelligent Systems*, 2025) and HAM (Hard AE Mining; *IEEE TIFS*, 2025) on CIFAR10 using ResNet-18 model, evaluated using AutoAttack under perturbation budget $\epsilon = 8/255$. Results are reported as mean ± standard deviation across five random seeds.

| Method | AvgC ↑ | WstC ↑ | AvgR ↑ | WstR ↑ |
|---|---|---|---|---|
| TRIX | **82.25 ± 0.25** | **69.06 ± 0.68** | **49.03 ± 0.23** | **31.72 ± 1.16** |
| TRIX (EMA, $\alpha = 0.95$) | 82.07 ± 0.34 | 68.04 ± 1.37 | 47.28 ± 0.29 | 30.00 ± 1.71 |
| CODAT | 80.99 ± 0.16 | 66.64 ± 2.20 | 45.36 ± 0.41 | 29.60 ± 0.70 |
| HAM | 82.71 ± 0.22 | 68.66 ± 2.62 | 47.96 ± 0.65 | 27.88 ± 1.68 |

Figure 2: Average class-wise weights for all CIFAR-10 classes obtained using a ResNet-18 model trained with TRIX, aggregated over five random seeds. The results demonstrate a consistent pattern across runs: hard classes (e.g., cat, deer, bird) exhibit higher weights (approximately 1.10–1.24), whereas easy classes (e.g., automobile, airplane, ship, truck) maintain lower weights (approximately 0.84–0.97), with minimal variance across seeds. This consistency highlights the stability of the class-wise partitioning into easy and hard classes.

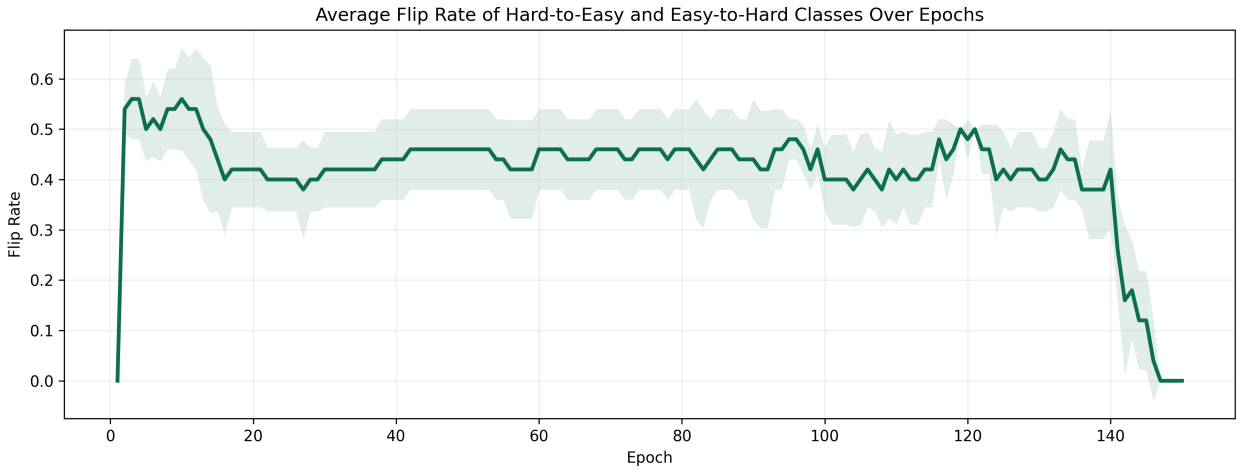



Figure 3: Average class-wise flip rate for CIFAR-10, computed using a ResNet-18 model trained with TRIX and aggregated over five random seeds. The flip rate measures how frequently classes transition between the easy and hard partitions during training. We observe that these transitions diminish over time and effectively converge in the final training phase. In particular, during epochs 141–150, there are negligible transitions between easy-to-hard and hard-to-easy classes, indicating stable and converged class partitioning.

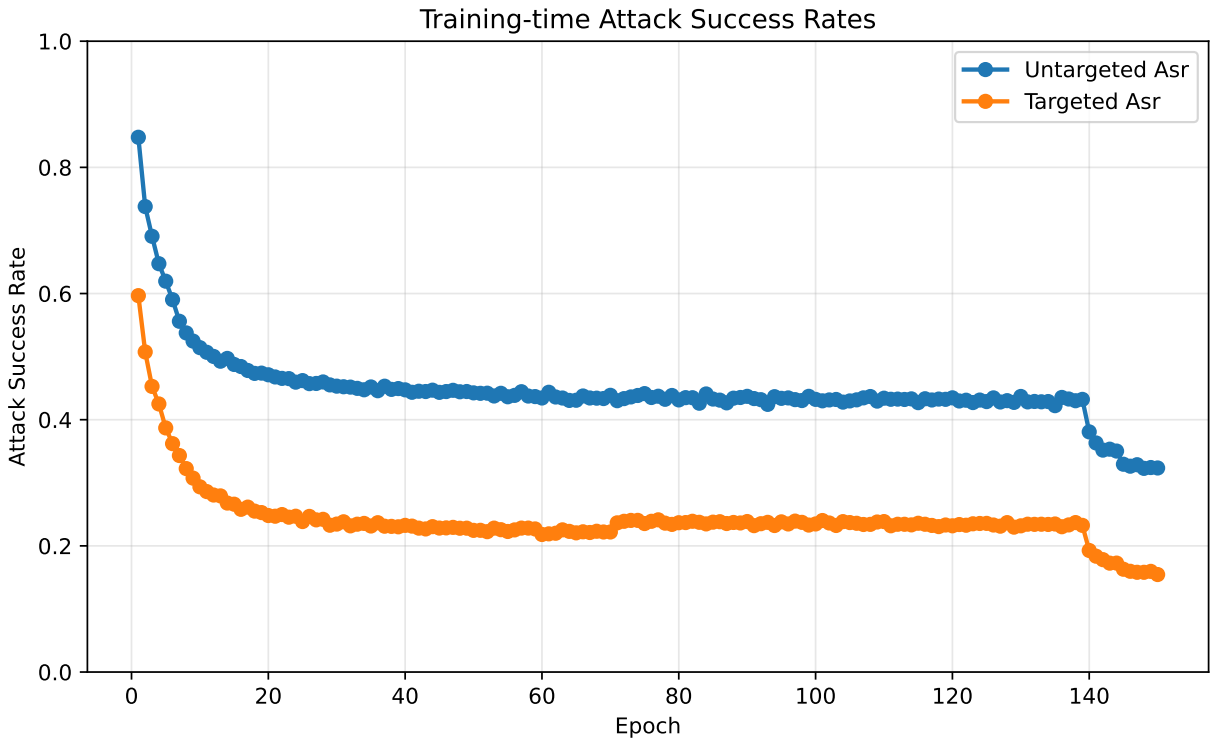



Figure 4: Average Attack Success rates of targeted and untargeted attacks during training of for all CIFAR-10 classes obtained using a ResNet-18 model trained with TRIX. we observe that attack success rates of targeted adversaries is lower than untargeted adversaries, targeted attacks are less successful in triggering misclassifications, but increase the diversity in false positive classes and thus specificity.

Table 3: AutoAttack evaluation on ResNet-18 for robust fairness approaches ($\epsilon = 8/255$).

CIFAR-10 (Best Checkpoint)

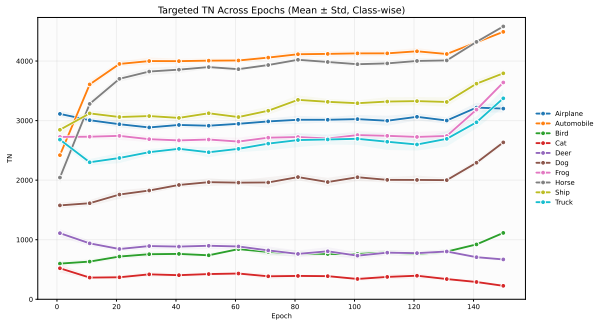
| Method | AvgC | WstC | AvgR | WstR |
|---|---|---|---|---|
| TRADES | – | – | – | – |
| + FRL | – | – | – | – |
| + BAT | – | – | – | – |
| + WAT | – | – | – | – |
| + CFA | – | – | – | – |
| + FAIRTAT | – | – | – | – |
| + DAFA | – | – | – | – |
| + ABSLD | – | – | – | – |
| + TRIX | – | – | – | – |

CIFAR-10 (Last Checkpoint)

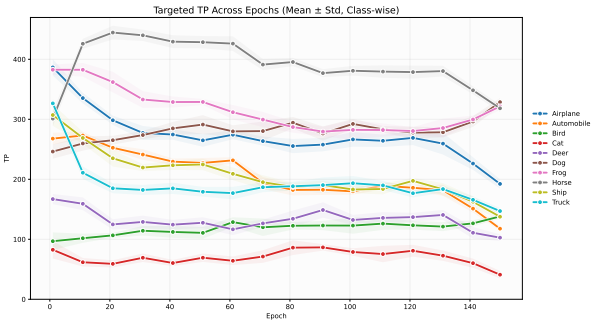
| Method | AvgC | WstC | AvgR | WstR |
|---|---|---|---|---|
| TRADES | – | – | – | – |
| + FRL | – | – | – | – |
| + BAT | – | – | – | – |
| + WAT | – | – | – | – |
| + CFA | – | – | – | – |
| + FAIRTAT | – | – | – | – |
| + DAFA | – | – | – | – |
| + ABSLD | – | – | – | – |
| + TRIX | – | – | – | – |

Table 4: AutoAttack evaluation on ResNet-10 for CIFAR-10 ($\epsilon = 8/255$). We report results when TRIX is applied to existing robust fairness approaches.

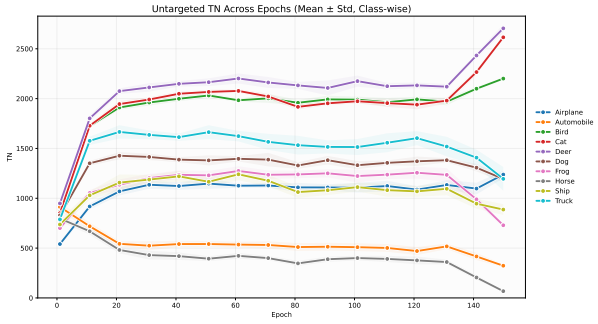
| Method | AvgC | WstC | AvgR | WstR |
|---|---|---|---|---|
| FRL | – | – | – | – |
| FRL + TRIX | – | – | – | – |
| BAT | – | – | – | – |
| BAT + TRIX | – | – | – | – |
| DAFA | – | – | – | – |
| TRIX | – | – | – | – |



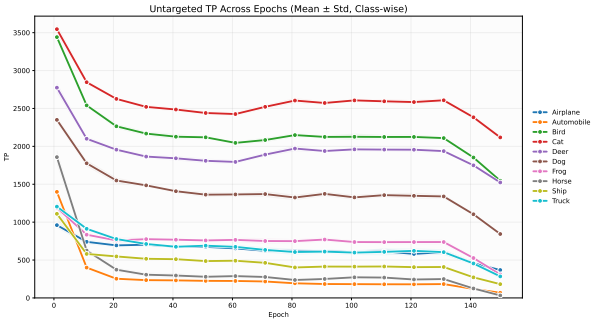

(a) True Negatives of Targeted Adversarial during training of TRIX




(b) True Positives of Targeted Adversarial during training of TRIX




(c) True Negatives of Untargeted Adversarial during training of TRIX




(d) True Positives of Untargeted Adversarial during training of TRIX

Figure 5: Training learning curves of TP/TN for targeted and untargeted adversarial training

Table 5: AutoAttack evaluation on ImageNet ($\epsilon = 4/255$) using a Swin-B adversarially fine-tuned model.

| Method | AvgC ↑ | WstC ↑ | AvgR ↑ | WstR ↑ |
|---|---|---|---|---|
| TRIX | 72.50 | 49.03 | 36.32 | 3 |
| DAFA | 72.49 | 48.05 | 36.98 | 2 |
| TRADES | 72.03 | 47.80 | 37.60 | 0 |